LumaNet: An innovative low-cost mesh network solution to uplift regions with little to no internet stability

Aryan Ruia and Nihal Gorthi

Introduction and Problem

Approximately 3.45 billion people, or about 43% of the global population, do not use mobile internet due to cost, lack of devices, skills, and infrastructure limitations, resulting in significant barriers to healthcare, education, digital inclusion, and economic advancement.

This investigation gauged the applicability of LumaNet, our decentralized offline mesh network, to reliably store, process, and provide stable, localized access to digital information in regions with unreliable connectivity at a lower cost and higher resilience than existing infrastructure. We hypothesized that multiple low-cost nodes in a mesh-network system can provide stable, localized access to information because the decentralized structure allows for effective data sharing without constant internet connectivity.

To test this, we engineered prototype nodes integrating Raspberry Pi computers with a dual-radio architecture: 900 MHz radios for long-range backbone communication and 2.4 GHz radios for local device access, allowing users to connect via standard Wi-Fi while data relays over the long-range network.

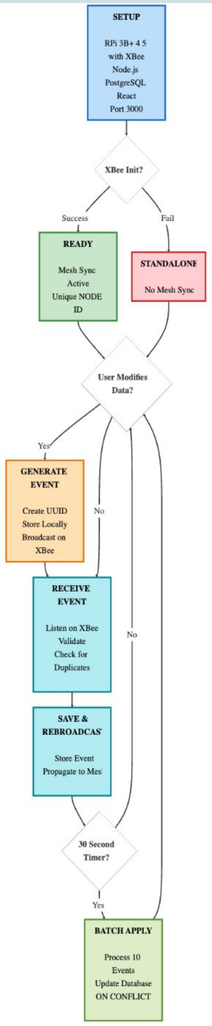

Figure 1. *LumaNet Mesh Architecture**

Methodology

We built three nodes using Raspberry Pi 3 B+, 5, and 4, each with a regulated XBee breakout board wired to GPIO pins (5 V, GND, TXD, RXD). A manual toggle switch allowed switching between the XBee 3 Pro (2.4 GHz, ~1200 m range) and XBee Pro 900 HP (900 MHz, extended range). Each node ran Raspberry Pi OS 64-bit Bookworm with Node.js 20.x, PostgreSQL 15.x, TypeScript 5.x, React 18.x, and Avahi for local network discovery.

As a proof of concept, we implemented LumaNet as a decentralized, offline-first LMS, with each node running an independent server, identical PostgreSQL databases, and Express REST APIs. The React frontend was served on port 3000, and each database included tables for users, subjects, student progress, and a sync_events table logging all modifications with UUIDs, timestamps, source node IDs, and JSONB payloads.

The system used event-sourcing for mesh data syncing. User-created or modified data generated sync events stored locally and broadcast via XBee serial at 115,200 baud using newline-delimited JSON (256-byte max). Receiving nodes validated, deduplicated, saved, and processed unapplied events every 30 seconds in batches of 10 with idempotent ON CONFLICT clauses, then rebroadcast events (excluding their own) to propagate changes. XBee modules were in API Mode with identical PAN IDs, each node had a unique NODE_ID, and if initialization failed, the backend ran standalone. Avahi provided local access at http://lumanet-node-X.local:3000.

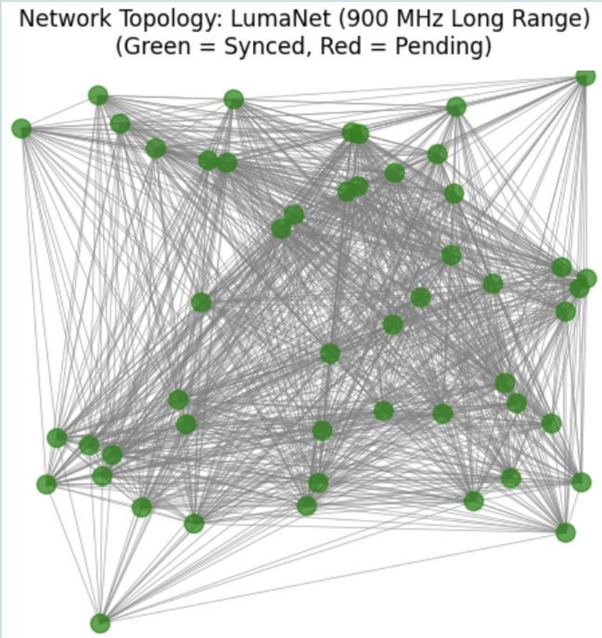

Figure 2. *Network Topology in a Single Simulation Run**

Results

Using a custom simulation, we evaluated the LumaNet decentralized mesh network consisting of 50 nodes across a 10 km x 10 km area. Utilizing the XBee 900HP radio configuration, the simulation modeled the synchronization of a 5 MB file—a scenario representative of decentralized educational content distribution. We found that this mid-sized mesh network achieves throughput within one order of magnitude of traditional solutions like 4G or satellite, while costing approximately 1% of traditional infrastructure.

The system achieved 100% synchronization efficiency in 359 seconds, with an average throughput of 114.1 kbps and an average node latency of 329.9 seconds. The simulated network was densely connected with 1,008 edges and an average node degree of 40.32. However, the high density led to a 74.9% reliability rate, as over 6,000 packets were lost due to simulated network collisions.

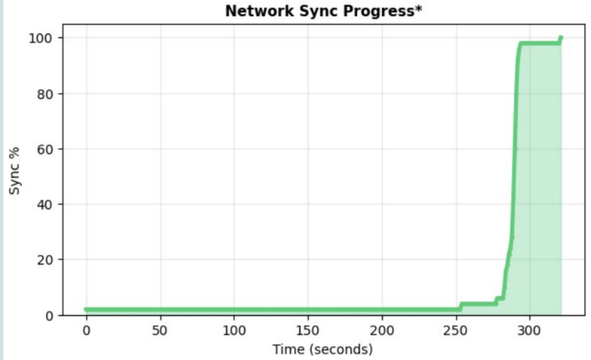

Figure 3. *Graph of Sync Progress over Time**

Energy efficiency was a highlight of the prototype. The entire synchronization consumed roughly 0.02 Wh, meaning battery depletion was nearly non-existent. With an idle power draw of only 15 mW and a transmission power of 950 mW, the nodes are well-suited for long-term solar or battery operation without significant maintenance or frequent charging.

The financial comparison revealed that LumaNet's initial cost ($8,750) is 132.1 times less expensive than a fiber-optic alternative ($1.15 million). When accounting for fiber's estimated annual maintenance of over $230,000, LumaNet offers projected total savings of $2.3 million. This results in a 5-year payback period, establishing LumaNet as a robust, low-power, and highly cost-effective solution for rural digital inclusion.

Discussion and Future Work

The experimental findings confirm our hypothesis that a decentralized mesh network can provide stable, localized digital access at a fraction of traditional infrastructure costs. With a 100% synchronization rate achieved in under six minutes for educational content distribution, LumaNet presents a viable alternative for regions where fiber-optic cables are economically or geographically prohibitive. The massive cost disparity (deploying LumaNet at 1/132nd the cost of fiber) underscores its potential for rapid, scalable deployment in underserved communities.

However, the 74.9% reliability rate observed in the 50-node simulation indicates a critical bottleneck in high-density scenarios. While the network remained fully connected with a high average node degree of 40.32 , the significant volume of lost packets due to simulated collisions suggests that the current "gossip" synchronization protocol requires refinement as the node count increases.

The system's negligible energy footprint (0.02 Wh total) validates its suitability for off-grid operation. With an idle power draw of only 15 mW, these nodes could operate indefinitely using small solar panels or standard battery packs without requiring significant maintenance.

Future Work will focus on:

- Scalability Testing: Expand the simulation to 100+ nodes to determine the XBee 900HP's throughput limits under extreme congestion.
- Hybrid Radio Logic: Implement dynamic frequency switching between 900 MHz and 2.4 GHz to balance long-range reliability with high-speed local access based on real-time load.
- Enhanced Security: Integrate advanced encryption protocols to protect sensitive data.
- Internet Bridging: Sync data to the cloud via Wi-Fi when available, enabling global updates while preserving offline operation.

References

Achieving universal connectivity by 2030. (n.d.). United Nations. Retrieved October 2, 2025, from https://www.un.org/digital-emerging-technologies/content/global-connectivity
Cleaning up electronic waste (E-Waste). (2025, May 16). EPA. Retrieved October 2, 2025, from https://www.epa.gov/international-cooperation/cleaning-electronic-waste-e-waste
Individuals using the internet (% of population). (n.d.). World Bank Group. Retrieved October 2, 2025, from https://data.worldbank.org/indicator/IT.NET.USER.ZS
Johnson, H. (2024, September 25). *About 2.5 billion people lack internet access: How connectivity can unlock their potential*. World Economic Forum. Retrieved October 2, 2025, from https://www.weforum.org/stories/2024/09/2-5-billion-people-lack-internet-access-how-connectivity-can-unlock-their-potential/
New GSMA report shows mobile internet connectivity continues to grow globally but barriers for 3.45 billion unconnected people remain. (2024, October). GSMA. Retrieved October 2, 2025, from https://www.gsma.com/newsroom/press-release/new-gsma-report-shows-mobile-internet-connectivity-continues-to-grow-globally-but-barriers-for-3-45-billion-unconnected-people-remain/
31 percent of rural households lack access to broadband internet, according to white paper. (2018, October 8). Center for Rural Affairs. Retrieved October 2, 2025, from https://www.cfra.org/news-release/31-percent-rural-households-lack-access-broadband-internet-according-white-paper
Used household batteries. (2025, January 7). EPA. Retrieved October 2, 2025, from https://www.epa.gov/recycle/used-household-batteries
Winslow, J. (2019, July 26). *America's digital divide*. Pew. Retrieved October 2, 2025, from https://www.pew.org/en/trust/archive/summer-2019/americas-digital-divide?
*All images generated by student researchers